No rational point of order 45 on a geometrically split abelian surface simple over $\mathbb{Q}$

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Theorem. *Let $A/\mathbb{Q}$ be a principally polarized abelian surface that is simple over $\mathbb{Q}$ and geometrically split. Then $A(\mathbb{Q})$ has no point of order 45.*

Corollary. *No genus-2 Jacobian over $\mathbb{Q}$ that is geometrically split and simple over $\mathbb{Q}$ has a rational torsion subgroup containing an element of order 45. In particular [45] does not occur as the rational torsion subgroup of such a Jacobian.*

Here "geometrically split" means that $A_{\bar{\mathbb{Q}}}$ is isogenous to a product of two elliptic curves. We impose no restriction on the degree of this isogeny, on the field of definition of the geometric endomorphisms or on the integral endomorphism ring. The principal polarization and the simplicity over $\mathbb{Q}$ are used only in Section 4. The theorem says nothing about surfaces split over $\mathbb{Q}$: a product of elliptic curves over $\mathbb{Q}$ with rational points of orders 5 and 9 has a rational point of order 45, and the census contains Jacobians split over $\mathbb{Q}$ with torsion subgroup [45]. For geometric squares we prove a stronger statement, which needs neither simplicity over $\mathbb{Q}$ nor a polarization:

Proposition A. Let $A/\mathbb{Q}$ be an abelian surface such that $A_{\bar{\mathbb{Q}}}$ is isogenous to the square of an elliptic curve. Then $A(\mathbb{Q})$ has no point of order 45.

Outline. Section 2 proves two lemmas on elliptic curves: coprime torsion can be combined in an isogeny class, and when $E[p]^{\mathrm{ss}}$ is a sum of two distinct characters, every Galois-stable lattice in $V_p(E)^r$ splits into lattices of rank 2. Section 3 proves Proposition A: CM squares by a point count over $\mathbb{F}_2$, non-CM squares by the residual-representation lemmas of [P63] at the primes 5 and 3 and, when the endomorphism field is quadratic, by moving rational 9-torsion to an elliptic curve over that field. In the remaining case $A_{\bar{\mathbb{Q}}}$ is isogenous to $E \times F$ with E and F not isogenous. Then E and F are conjugate over a quadratic field K, and the principal polarization identifies each of them with the quotient of A_K by the other. Section 4 shows that $E(K)_{\mathrm{tors}} \cong \mathbb{Z}/15$, that A_K is an (n,n)-gluing of E and E^σ with $v_3(n) = 1$, and that its 3-part is a conjugate $(3,3)$-gluing B over $\mathbb{Q}$ with a rational point of order 9. Section 5 turns this into two conditions on E: the norm of its discriminant is a rational cube, and its flex coefficient a_3 satisfies $a_3/\sigma(a_3) \in K^{\times 3}$. Section 7 shows that no quadratic point of $X_1(15)$ satisfies both. On each of the four families of quadratic points, the second condition forces a primitive value of a cyclotomic binary quartic to be a cube or five times a cube; a descent in $\mathbb{Q}(\zeta_5)$ following Poonen and elliptic Chabauty over $\mathbb{Q}(\zeta_5)$ (Section 6) leave four parameter values, where the point is a cusp or the first condition fails.

1 Conventions and inputs

All number fields are subfields of a fixed algebraic closure $\bar{\mathbb{Q}}$. For a number field k we write $G_k = \mathrm{Gal}(\bar{\mathbb{Q}}/k)$, and $\bar{\chi}_\ell$ for the mod-ℓ cyclotomic character. For an abelian variety X,

$T_\ell(X)$ is the Tate module, $V_\ell(X) = T_\ell(X) \otimes \mathbb{Q}_\ell$, $X[n]$ is the kernel of multiplication by n and $\mathrm{End}^0 = \mathrm{End} \otimes \mathbb{Q}$. For an elliptic curve E, e_n is the Weil pairing on $E[n]$; an *anti-isometry* $E[n] \to E'[n]$ is an isomorphism ψ with $e_n(\psi x, \psi y) = e_n(x,y)^{-1}$. For a quadratic field K we write σ for its nontrivial automorphism and $\tilde{\sigma} \in G_{\mathbb{Q}}$ for a lift of σ; for a variety Y over K, Y^σ is the conjugate variety and $\sigma : Y(\bar{\mathbb{Q}}) \to Y^\sigma(\bar{\mathbb{Q}})$ the bijection induced by $\tilde{\sigma}$ [P2214, Section 1]. We write $k^{\times 3}$ for the nonzero cubes of a field k, and v_ℓ for the ℓ-adic valuation. We use the functoriality of the Weil pairing and Galois descent for morphisms of varieties without further comment.

- **(I1) Torsion of elliptic curves over $\mathbb{Q}$ and over quadratic fields.** By Mazur [Ma, Theorem (8)], no elliptic curve over $\mathbb{Q}$ has a rational point of order 15. By Kenku–Momose [KM, Theorem (0.1)] and Kamienny [Ka] (see the statement in [Na, Section 1]), the torsion subgroup of an elliptic curve over a quadratic field is isomorphic to $\mathbb{Z}/m$ with $1 \le m \le 16$ or $m = 18$, to $\mathbb{Z}/2 \times \mathbb{Z}/2m$ with $1 \le m \le 6$, to $\mathbb{Z}/3 \times \mathbb{Z}/3m$ with $m \in \{1, 2\}$, or to $\mathbb{Z}/4 \times \mathbb{Z}/4$. In particular there is no point of order 45, and a torsion subgroup containing a point of order 15 is isomorphic to $\mathbb{Z}/15$. Over $\mathbb{Q}(\sqrt{-3})$ only Mazur's groups, $\mathbb{Z}/3 \times \mathbb{Z}/3$ and $\mathbb{Z}/3 \times \mathbb{Z}/6$ occur [Na, Theorem 2(ii)]; in particular there is no point of order 15. (The finite checks in this item are part A of the certificate.)

- **(I2) Reduction** [ST], as stated in [P63, Section 1]: let K' be a finite extension of $\mathbb{Q}_p$ with residue field k and X an abelian variety over K'. (a) Good reduction is detected by the inertia action on $T_\ell(X)$, $\ell \neq p$ [ST, §1, Theorem 1]; in particular it is preserved by isogenies [ST, §1, Corollary 2]. (b) If X has good reduction $\widetilde{X}$ and m is prime to p, then $X(K')[m]$ injects into $\widetilde{X}(k)$ [ST, Lemma 2]. (c) If X has potentially good reduction, then it has good reduction over a finite extension of K' with residue field k [ST, §2]. (d) (Weil.) An abelian surface Y over $\mathbb{F}_q$ has at most $(1 + \sqrt{q})^4$ points.

- **(I3) The case distinction** [P63, Sections 2 and 5]. A geometrically split abelian surface $A/\mathbb{Q}$ is of exactly one of the types **(N)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to $E_1 \times E_2$ with E_1, E_2 not isogenous), **(CM)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to E^2 with E a CM curve) and **(S)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to E^2 with E without CM). In case (S), $D = \mathrm{End}^0(A_{\bar{\mathbb{Q}}}) \cong M_2(\mathbb{Q})$; L is the smallest field over which all geometric endomorphisms are defined; $\Gamma = \mathrm{Gal}(L/\mathbb{Q})$ embeds in $\mathrm{PGL}_2(\mathbb{Q})$, the element g acting on D by conjugation with $R_g \in D^\times$; and Γ is one of C_1, C_2, C_3, C_4, C_6, V_4, D_6, D_8, D_{12} (D_{2m} is dihedral of order $2m$). An element of Γ of order $m \in \{3, 4, 6\}$ has a unique representative R with $\mathrm{tr}\, R = 1$, and $R^2 - R + 1/c = 0$ with $c = 1, 2, 3$ respectively.

- **(I4)** [P63, Lemma 2]. Let X be an abelian surface over a field k, $C \subset X$ an elliptic curve defined over k, and $Q \in X(k)$ of prime order q. Then either $Q \in C(k)$, or the image of Q in $(X/C)(k)$ has order q.

- **(I5)** [P63, Lemma 3]. In case (S), every elliptic curve $C \subset A_{\bar{\mathbb{Q}}}$ is defined over L, any two of them are L-isogenous, and A_L/C is L-isogenous to C.

- **(I6) Residual representations** [P63, Section 6]. Let A be of type (S), p an odd prime with $A(\mathbb{Q})[p] \neq 0$, and $D_p = D \otimes \mathbb{Q}_p$.

  (a) (*Unramified order*, [P63, Lemma 6].) Suppose that a $\mathbb{Z}_p$-order $\mathcal{B} \subset D_p$, isomorphic to the ring of integers of the unramified quadratic extension of $\mathbb{Q}_p$, is preserved by Γ, and that some $g_0 \in \Gamma$ has a representative $R_{g_0} \in \mathcal{B}^\times$ whose reduction in $\mathcal{B}/p\mathcal{B} \cong \mathbb{F}_{p^2}$ is not in $\mathbb{F}_p$. Then $\mathbb{Q}(\zeta_p) \subseteq L$.

(b) (*Split blocks*, [P63, Lemma 7].) Suppose that $\mathcal{B} = \mathbb{Z}_p e_1 \oplus \mathbb{Z}_p e_2 \subset D_p$, with orthogonal idempotents e_1, e_2 of rank one and $e_1 + e_2 = 1$, is preserved by Γ, and that some element of Γ exchanges e_1 and e_2. Let Γ_0 be the stabilizer of e_1 and $K_0 = L^{\Gamma_0}$. If the character η by which G_{K_0} acts on the line $e_1 D_p e_2$ has nontrivial reduction modulo p, then $\mathbb{Q}(\zeta_p) \subseteq L$.

In (a) the proof replaces $T_p(A)$ by the lattice generated over the Γ-stable maximal order $\mathrm{End}_{\mathbb{Z}_p}(\Lambda) \supset \mathcal{B}$, where $\Lambda \subset \mathbb{Q}_p^2$ is a $\mathcal{B}$-stable lattice, and in (b) by $\mathcal{B}\,T_p(A)$. Both proofs end with the polarization duality of [P63, Lemma 5] ([P30, Lemma 3]), which needs only a polarization of A over $\mathbb{Q}$, of any degree. The proofs use nothing else about A: neither the simplicity over $\mathbb{Q}$ nor the point of order 63 assumed in [P63, Section 2]. The same arguments appear in [P30, Lemma 4 and Section 8].

- **(I7) Weil restriction** [Mi, §1]. Let K be a quadratic field and E an elliptic curve over K. The Weil restriction $\mathrm{Res}_{K/\mathbb{Q}} E$ is an abelian surface over $\mathbb{Q}$ with $(\mathrm{Res}_{K/\mathbb{Q}} E)(\mathbb{Q}) = E(K)$ and $(\mathrm{Res}_{K/\mathbb{Q}} E)_K \cong E \times E^\sigma$. Its $\bar{\mathbb{Q}}$-points are the pairs (x, y) with $x \in E(\bar{\mathbb{Q}})$ and $y \in E^\sigma(\bar{\mathbb{Q}})$; an element $g \in G_K$ maps (x, y) to (gx, gy), an element $g \notin G_K$ maps it to (gy, gx), and the rational points are the pairs $(P, \sigma P)$ with $P \in E(K)$.

- **(I8) Complements and gluing** [BL, Section 5.3 and Corollary 12.1.5], [FK], [Ka97]. For an elliptic curve C in an abelian surface X with a polarization λ, the complement of C is the connected component of the kernel of $i_C^\vee \circ \lambda$, where i_C is the inclusion. A principally polarized abelian surface containing two complementary elliptic curves E, F is the quotient of $E \times F$ by the graph of an anti-isometry $E[n] \to F[n]$ (an (n, n)-gluing). We prove the parts of this that we use in Section 4.

- **(I9) Tate normal form** [Ku, Table 3, the entry for $\mathbb{Z}/5$]. Let k be a field of characteristic 0 and put $E_d : y^2 + (1-d)xy - dy = x^3 - dx^2$, whose discriminant is $d^5(d^2 - 11d - 1)$. Every pair (E, P) of an elliptic curve over k and a point $P \in E(k)$ of order 5 is isomorphic over k to $(E_d, (0, 0))$ for some $d \in k$ with $d^5(d^2 - 11d - 1) \neq 0$. Kubert works over $\mathbb{Q}$, but his derivation from the Tate normal form uses only field operations.

2 Two lemmas on elliptic curves

Lemma 1 (combining coprime torsion). Let k be a number field and m, n coprime positive integers. If a k-isogeny class of elliptic curves over k contains a curve with a k-rational point of order m and a curve with a k-rational point of order n, then it contains a curve with a k-rational point of order mn.

Proof. This is [P63, Lemma 1]. Let $f : E_m \to E_n$ be a k-isogeny between the two curves and H the subgroup of $\ker f$ of elements whose order involves only primes dividing n; it is characteristic, hence defined over k, and $f = b \circ a$ with $a : E_m \to E' = E_m/H$. As $\deg a$ is prime to m, a maps the point of order m to a point of order m in $E'(k)$; as $\deg b$ is prime to n, the dual isogeny $\hat{b}$ maps the point of order n to a point of order n in $E'(k)$. Their sum has order mn. □

Lemma 2 (stable lattices). Let k be a number field, p a prime, and E/k an elliptic curve such that $E[p]^{\mathrm{ss}} \cong \alpha \oplus \beta$ for two distinct characters $\alpha, \beta : G_k \to \mathbb{F}_p^\times$. Put $T = T_p(E)$ and $V = V_p(E)$, and let $r \geq 1$.

(a) Every G_k-stable $\mathbb{Z}_p$-lattice in V^r is a direct sum of r G_k-stable lattices of rank 2, each contained in a G_k-subrepresentation of V^r isomorphic to V.

(b) If an abelian variety Y over k is k-isogenous to E^r, then there are elliptic curves $E_1, \dots, E_r$ over k, each k-isogenous to E, such that $Y[p^\infty] \cong \bigoplus_j E_j[p^\infty]$ as G_k-modules. In particular, if $Y(k)$ has a point of order p^e, then so has some $E_j(k)$.

Proof. Let $\mathcal{R}$ be the $\mathbb{Z}_p$-span of the image of G_k in $\mathrm{End}_{\mathbb{Z}_p}(T) \cong M_2(\mathbb{Z}_p)$. It is a finitely generated, hence free, $\mathbb{Z}_p$-submodule of $\mathrm{End}_{\mathbb{Z}_p}(T)$, and a subring containing 1. Choose $g \in G_k$ with $\alpha(g) \neq \beta(g)$. The characteristic polynomial of g on T has the distinct roots $\alpha(g)$, $\beta(g)$ modulo p, so by Hensel's lemma it has roots $\lambda_1, \lambda_2 \in \mathbb{Z}_p$ with $\lambda_1 - \lambda_2 \in \mathbb{Z}_p^\times$. By the Cayley–Hamilton theorem, $\epsilon_1 = (g - \lambda_2)/(\lambda_1 - \lambda_2)$ and $\epsilon_2 = 1 - \epsilon_1$ are orthogonal idempotents, and they lie in $\mathcal{R}$. They have rank one, since g is not a scalar, so $T = \mathbb{Z}_p v_1 \oplus \mathbb{Z}_p v_2$ with $\epsilon_i T = \mathbb{Z}_p v_i$. In this basis $\epsilon_i \mathcal{R} \epsilon_i$ contains ϵ_i and is contained in $\mathbb{Z}_p \epsilon_i$, and the corners $\epsilon_1 \mathcal{R} \epsilon_2$, $\epsilon_2 \mathcal{R} \epsilon_1$ are $\mathbb{Z}_p$-submodules of the off-diagonal lines. As $\epsilon_1 + \epsilon_2 = 1$, $\mathcal{R}$ is the sum of its four corners, so

$$\mathcal{R} = \begin{pmatrix} \mathbb{Z}_p & p^a\mathbb{Z}_p \\ p^b\mathbb{Z}_p & \mathbb{Z}_p \end{pmatrix} \qquad \text{with } a, b \in \{0, 1, 2, \dots\} \cup \{\infty\},\ p^\infty = 0.$$

(a) Write $V^r = V \otimes_{\mathbb{Q}_p} U$ with $U = \mathbb{Q}_p^r$, where G_k acts on the first factor. A G_k-stable lattice $T' \subset V^r$ is stable under $\mathcal{R} \otimes 1$, so $T' = \epsilon_1 T' \oplus \epsilon_2 T'$ with $\epsilon_i T' = v_i \otimes \Lambda_i$ for lattices $\Lambda_1, \Lambda_2 \subset U$. The off-diagonal elements of $\mathcal{R}$ give $p^a \Lambda_2 \subseteq \Lambda_1$ and $p^b \Lambda_1 \subseteq \Lambda_2$, and conversely every lattice $v_1 \otimes \Lambda_1 \oplus v_2 \otimes \Lambda_2$ with these two inclusions is $\mathcal{R}$-stable, hence G_k-stable. By the elementary divisor theorem there are a basis $w_1, \dots, w_r$ of Λ_1 and integers $d_1, \dots, d_r$ with $\Lambda_2 = \bigoplus_j p^{d_j} \mathbb{Z}_p w_j$. The two inclusions say $a + d_j \geq 0$ and $d_j \leq b$ for all j, so each $T'_j = \mathbb{Z}_p(v_1 \otimes w_j) \oplus p^{d_j}\mathbb{Z}_p(v_2 \otimes w_j)$ is G_k-stable, and $T' = \bigoplus_j T'_j$ with $T'_j \subset V \otimes w_j \cong V$.

(b) An isogeny identifies $V_p(Y)$ with V^r G_k-equivariantly, and $T_p(Y)$ with a lattice $T' = \bigoplus_j T'_j$ as in (a). Identify $V \otimes w_j$ with V and choose $c_j \geq 0$ with $p^{c_j} T'_j \supseteq T$. Then $C_j = p^{c_j} T'_j / T$ is a finite G_k-stable subgroup of $V/T = E[p^\infty]$, and $E_j = E/C_j$ is an elliptic curve over k, k-isogenous to E, with $T_p(E_j) = p^{c_j} T'_j \cong T'_j$. Hence $Y[p^\infty] = V_p(Y)/T_p(Y) \cong \bigoplus_j (V \otimes w_j)/T'_j \cong \bigoplus_j E_j[p^\infty]$. Taking G_k-invariants, $Y(k)[p^\infty] \cong \bigoplus_j E_j(k)[p^\infty]$, and a point of order p^e in the sum has a coordinate of order p^e. □

The lemma needs neither the irreducibility of V nor any hypothesis on complex multiplication: only the two idempotents ϵ_1, ϵ_2 of $\mathcal{R}$ are used. They exist because $\alpha \neq \beta$; this is why the field $\mathbb{Q}(\sqrt{-3})$ has to be excluded in the case $\Gamma = C_2$ of Section 3.

3 The geometric-square case

Proof of Proposition A. Let $A_{\bar{\mathbb{Q}}}$ be isogenous to E^2, and suppose that $R_{45} \in A(\mathbb{Q})$ has order 45. Put $R_9 = 5R_{45}$, $R_5 = 9R_{45}$ and $R_3 = 15R_{45}$, of orders 9, 5 and 3.

The CM case. Suppose that E has complex multiplication. As in [P63, Section 4], E has potentially good reduction at 2, because its j-invariant is an algebraic integer [ST, §2]; so $A_{\mathbb{Q}_2}$ acquires good reduction over a finite extension by (I2)(a), and by (I2)(c) over a finite extension K' of $\mathbb{Q}_2$ with residue field $\mathbb{F}_2$. By (I2)(b) the subgroup generated by R_{45}, of odd order, injects into the reduction, whose number of points is at most $(1+\sqrt{2})^4 = 17 + 12\sqrt{2} < 34$ by (I2)(d). This contradicts $45 \leq 34$. ([P30, Proposition 2] shows more: $A(\mathbb{Q}_2)$ has no point of order 15.)

The non-CM case. Now A is of type (S), with D, L, Γ as in (I3). Since R_5 and R_3 are rational, (I6) applies with $p = 5$ and with $p = 3$. Modulo 5 the discriminants -3, -1, $-\frac{1}{3}$ of the polynomials $X^2 - X + \frac{1}{c}$ of (I3) are 2, $4 = 2^2$ and 3; the nonzero squares modulo 5 are 1 and 4. The field $\mathbb{Q}(\zeta_5)$ is cyclic of degree 4, so if $\mathbb{Q}(\zeta_5) \subseteq L$, then Γ has a quotient isomorphic to C_4. We go through the groups of (I3).

(i) $\Gamma = C_1$. Then $L = \mathbb{Q}$, and by (I5) every elliptic curve $C \subset A$ is defined over $\mathbb{Q}$, with A/C $\mathbb{Q}$-isogenous to C. By (I4), applied to R_3 and R_5, the $\mathbb{Q}$-isogeny class of C contains curves with rational points of orders 3 and 5, and Lemma 1 gives an elliptic curve over $\mathbb{Q}$ with a rational point of order 15, contrary to (I1).

(ii) $\Gamma \in \{C_3, C_6, D_6, D_{12}\}$. Let r be a rotation of order 3 or 6 generating the rotation subgroup and R its trace-one representative, with $c \in \{1, 3\}$. Every element of Γ maps R to R or $1 - R$ [P63, Section 7], so $\mathcal{B} = \mathbb{Z}_5[R]$ is preserved by Γ. Its discriminant is a non-square unit modulo 5, so $\mathcal{B}$ is the ring of integers of the unramified quadratic extension of $\mathbb{Q}_5$; $R \in \mathcal{B}^\times$ because $\det R = 1/c$ is a unit, and its reduction is a root of an irreducible quadratic over $\mathbb{F}_5$. By (I6)(a), $\mathbb{Q}(\zeta_5) \subseteq L$. But C_3, C_6 and the abelianizations C_2 and $C_2 \times C_2$ of D_6 and D_{12} have no quotient C_4.

(iii) $\Gamma = D_8$. Let r be a rotation of order 4 and R its trace-one representative, $R^2 - R + \frac{1}{2} = 0$. Again $\mathcal{B} = \mathbb{Z}_5[R]$ is preserved by Γ. The discriminant $-1 \equiv 2^2$ is a square modulo 5, so the roots λ_1, λ_2 of $X^2 - X + \frac{1}{2}$ lie in $\mathbb{Z}_5^\times$, with $\lambda_1 - \lambda_2$ a unit, and $\mathcal{B} = \mathbb{Z}_5 e_1 \oplus \mathbb{Z}_5 e_2$ with $e_1 = (R - \lambda_2)/(\lambda_1 - \lambda_2)$ and $e_2 = 1 - e_1$. The rotations fix e_1 and e_2, and the reflections, which map R to $1 - R$, exchange them because $\lambda_1 + \lambda_2 = 1$. The stabilizer of e_1 is the rotation subgroup, and $RXR^{-1} = (\lambda_1/\lambda_2)X$ for $X \in e_1 D_5 e_2$, so η maps a lift of r to λ_1/λ_2, a primitive fourth root of unity, whose reduction modulo 5 still has order 4. By (I6)(b), $\mathbb{Q}(\zeta_5) \subseteq L$; but the abelianization of D_8 is $C_2 \times C_2$.

(iv) $\Gamma = V_4$. The map $[M] \mapsto v_5(\det M) \bmod 2$ is a homomorphism $\Gamma \to \mathbb{Z}/2$; choose $r \neq 1$ in its kernel and a representative R of trace 0, scaled by a power of 5 so that $\det R \in \mathbb{Z}_5^\times$. Then $R^2 = a$ with $a \in \mathbb{Q}^\times \cap \mathbb{Z}_5^\times$. As Γ is abelian, every R_g conjugates R to $\pm R$, so $\mathcal{B} = \mathbb{Z}_5[R]$ is preserved by Γ. If a is not a square modulo 5, then $\mathcal{B}$ is the unramified quadratic order, $R \in \mathcal{B}^\times$, its reduction is not in $\mathbb{F}_5$, and (I6)(a) applies. If $a = b^2$ with $b \in \mathbb{Z}_5^\times$, then $\mathcal{B} = \mathbb{Z}_5 e_+ \oplus \mathbb{Z}_5 e_-$ with $e_\pm = (1 \pm R/b)/2$. Some element of Γ conjugates R to $-R$: otherwise Γ would embed in the image of the centralizer $\mathbb{Q}_5[R]^\times \cong \mathbb{Q}_5^\times \times \mathbb{Q}_5^\times$ in $\mathrm{PGL}_2(\mathbb{Q}_5)$, which is $\cong \mathbb{Q}_5^\times$ and has only cyclic finite subgroups. That element exchanges e_+ and e_-; the stabilizer of e_+ is $\{1, r\}$, and $RXR^{-1} = -X$ for $X \in e_+ D_5 e_-$, so η is the nontrivial quadratic character and (I6)(b) applies. In both cases $\mathbb{Q}(\zeta_5) \subseteq L$, which is impossible because L is biquadratic.

(v) $\Gamma = C_4$. Here we use the prime 3. Let r generate Γ and let R be its trace-one representative, $R^2 - R + \frac{1}{2} = 0$. The discriminant $-1 \equiv 2$ is not a square modulo 3, so $\mathcal{B} = \mathbb{Z}_3[R]$ is the ring of integers of the unramified quadratic extension of $\mathbb{Q}_3$; it is fixed by $\Gamma = \langle r \rangle$; $R \in \mathcal{B}^\times$ since $\det R = \frac{1}{2}$ is a 3-adic unit; and its reduction is a root of the irreducible polynomial $X^2 - X + 2$ over $\mathbb{F}_3$. By (I6)(a), $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3}) \subseteq L$. But a cyclic quartic field L does not contain $\mathbb{Q}(\sqrt{-3})$: complex conjugation would restrict to an element of $\mathrm{Gal}(L/\mathbb{Q}) \cong C_4$ that acts nontrivially on the unique quadratic subfield, hence to an element of order 4, whereas complex conjugation has order at most 2. (Equivalently, L is totally real by the argument of [P63, Section 9.1], which uses only that every element of Γ is the class of a power R^k with $\det R^k = 2^{-k} > 0$.)

(vi) $\Gamma = C_2$. Then $L = K$ is a quadratic field. By (I5), an elliptic curve $C \subset A_{\bar{\mathbb{Q}}}$ is defined over K, and A_K/C is K-isogenous to C; with a complement of C, also K-isogenous to C, we see that A_K is K-isogenous to C^2. By (I4), applied over K to R_3 and R_5, and by Lemma 1, the K-isogeny class of C contains a curve E_0 with a K-rational point of order 15. By (I1), $E_0(K)_{\mathrm{tors}} \cong \mathbb{Z}/15$ and $K \neq \mathbb{Q}(\sqrt{-3})$. The semisimplification of $E_0[3]$ is $\mathbf{1} \oplus \bar{\chi}_3$ (a rational point of order 3 and the Weil pairing), and $\bar{\chi}_3$ is nontrivial on G_K because $K \neq \mathbb{Q}(\sqrt{-3})$. As A_K is K-isogenous to E_0^2, Lemma 2(b) with $p = 3$

and $r = 2$, applied to the point $R_9 \in A(K)$ of order 9, gives an elliptic curve E_2, K-isogenous to E_0, with a K-rational point of order 9. Lemma 1, with E_2 and E_0, gives an elliptic curve over K with a K-rational point of order 45, contrary to (I1).

This covers all groups in (I3). □

No step uses simplicity over $\mathbb{Q}$, and the only polarization used is a polarization over $\mathbb{Q}$ of arbitrary degree, inside (I6).

4 Nonisogenous factors: the elliptic curves and the gluing

Lemma 3. Let $A/\mathbb{Q}$ be a principally polarized abelian surface that is simple over $\mathbb{Q}$ and of type (N), and suppose that $A(\mathbb{Q})$ has a point of order 45. Then there are a quadratic field K, an elliptic curve E over K and a point $P \in E(K)$ of order 3 with the following properties.

(a) $E(K)_{\mathrm{tors}} \cong E^\sigma(K)_{\mathrm{tors}} \cong \mathbb{Z}/15$.

(b) There is a G_K-equivariant anti-isometry $\psi : E[3] \to E^\sigma[3]$ with $\psi(P) = -\sigma(P)$.

(c) There are an abelian surface B over $\mathbb{Q}$, isogenies $\varphi : \mathrm{Res}_{K/\mathbb{Q}} E \to B$ and $\Phi : B \to \mathrm{Res}_{K/\mathbb{Q}} E$ over $\mathbb{Q}$ with $\Phi \circ \varphi = [3]$, such that $\ker\varphi$ is the graph of ψ (over K), and a point $R \in B(\mathbb{Q})$ of order 9 with $\Phi(R) = (P, \sigma P)$.

Proof. Let $\lambda : A \to A^\vee$ be the principal polarization, which is defined over $\mathbb{Q}$, and let $R_{45} \in A(\mathbb{Q})$ have order 45.

The two elliptic curves. As in [P63, Section 3], $A_{\bar{\mathbb{Q}}}$ contains exactly two elliptic curves, and $G_{\mathbb{Q}}$ permutes them. If it fixed them, they would be defined over $\mathbb{Q}$ and A would not be simple over $\mathbb{Q}$. So they are defined over a quadratic field K and exchanged by $\tilde\sigma$; call them E and $F = \tilde\sigma(E) = E^\sigma$.

Each is the quotient by the other. For an elliptic curve $C \subset A_K$ let $i_C : C \to A_K$ be the inclusion and $\lambda_C^0 : C \to C^\vee$ the canonical principal polarization. The dual $i_E^\vee : A_K^\vee \to E^\vee$ of the inclusion is surjective with connected kernel $(A_K/E)^\vee$, so, λ being an isomorphism, the kernel of $i_E^\vee \circ \lambda$ is a connected one-dimensional subgroup of A_K, the complement of E in (I8). It is not E, because $i_E^\vee \lambda i_E$ is a polarization of E, hence an isogeny. So it is F, the only other elliptic curve. Thus $\pi_E = (\lambda_E^0)^{-1} i_E^\vee \lambda : A_K \to E$ is surjective with kernel F, and induces $A_K/F \cong E$; in the same way $\pi_F = (\lambda_F^0)^{-1} i_F^\vee \lambda$ has kernel E and induces $A_K/E \cong F$.

The gluing. The polarization $i_E^\vee \lambda i_E$ of E is $n\lambda_E^0$ for some $n \geq 1$, with kernel $E[n]$; this kernel is also $E \cap \ker(i_E^\vee \lambda) = E \cap F$. Likewise $E \cap F = F[n']$ with $i_F^\vee \lambda i_F = n' \lambda_F^0$, and $n = n'$ by counting. So $E \cap F = E[n] = F[n]$ inside A_K. Let $\varphi_n : E \times F \to A_K$ be the addition map. Its kernel is $\{(x, -x) : x \in E \cap F\}$, the graph of $\psi_n : E[n] \to F[n]$, $x \mapsto -x$ (using $E[n] = F[n]$). The pull-back $\varphi_n^\vee \lambda \varphi_n$ is $n\lambda_0$ with $\lambda_0 = \lambda_E^0 \times \lambda_F^0$, because $i_E^\vee \lambda i_E = n\lambda_E^0$, $i_F^\vee \lambda i_F = n\lambda_F^0$ and $i_F^\vee \lambda i_E = 0$. Hence $\ker\varphi_n$ is isotropic for the pairing $(x, y) \mapsto e_n(x, \lambda_0 y)$ on $(E \times F)[n]$, which is the product of the Weil pairings of E and F: if $x, y \in \ker\varphi_n$ and $y = ny'$, then $\lambda_0 y = \varphi_n^\vee \lambda \varphi_n(y')$ with $\varphi_n(y') \in A_K[n]$, and $e_n(x, \varphi_n^\vee \lambda \varphi_n y') = e_n(\varphi_n x, \lambda \varphi_n y') = 1$. For $x, y \in E[n]$ this reads $e_n(x, y)\, e_n(\psi_n x, \psi_n y) = 1$: ψ_n is an anti-isometry. Put $\Phi_n = (\pi_E, \pi_F) : A_K \to E \times F$. For $x \in E$ and $y \in F$, $\Phi_n \varphi_n(x, y) = (\pi_E x, \pi_F y) = (nx, ny)$, so $\Phi_n \varphi_n = [n]$, and $\varphi_n \Phi_n = [n]$ because φ_n is surjective. Since λ is defined over $\mathbb{Q}$ and $\tilde\sigma(E) = F$, both maps commute with the action of $G_{\mathbb{Q}}$ of (I7) on $E \times F = E \times E^\sigma$, so they descend to isogenies $\varphi_n : \mathrm{Res}_{K/\mathbb{Q}} E \to A$ and $\Phi_n : A \to \mathrm{Res}_{K/\mathbb{Q}} E$ over $\mathbb{Q}$.

The torsion of $E(K)$. Let $G = \langle R_{45} \rangle$. Since $\tilde\sigma$ fixes G pointwise and maps E onto F, the subgroups $G \cap E$ and $G \cap F$ of the cyclic group G have the same order, hence coincide; call this subgroup H and put $d = \#H$. Then $E(K)$ contains H, of order d, and

$\pi_E(G) \cong G/(G \cap F) = G/H$, of order $45/d$. For $d \in \{1, 5, 9, 45\}$ this gives a point of order 45 in $E(K)$, contrary to (I1). Hence $d \in \{3, 15\}$, $E(K)$ has a point of order 15, and $E(K)_{\mathrm{tors}} \cong \mathbb{Z}/15$ by (I1); the same holds for $F = E^\sigma$. This is (a).

The sign. Put $R_9 = 5R_{45}$. As $d \in \{3, 15\}$, the point $P_0 = 3R_9$ of order 3 lies in $H \subseteq E \cap F = E[n]$; in particular $3 \mid n$. Let $P \in E(K)$ be P_0 regarded as a point of E. Since P_0 is rational, $P_0 = \tilde{\sigma}(P_0)$, and $\tilde{\sigma}(P_0)$ is the point $\sigma(P)$ of $F = E^\sigma$. So $\varphi_n(P, 0) = P_0 = \varphi_n(0, \sigma P)$, that is, $(P, -\sigma P) \in \ker \varphi_n$, and as $E(K)[3] = \langle P \rangle$,

$$\psi_n(Q) = -\sigma(Q) \qquad \text{for every } Q \in E(K)[3]. \tag{1}$$

With $+\sigma(Q)$ instead, condition (4) below would become $a_3\sigma(a_3) \in K^{\times 3}$; Remark 3 tests the sign on census curves.

The 3-adic valuation of n. Over K, $\ker \Phi_n = \varphi_n(E[n] \times F[n]) \cong (E[n] \times F[n])/\ker \varphi_n$, and $(U, V) \mapsto V - \psi_n(U)$ identifies it G_K-equivariantly with $F[n]$. Its K-rational points form $F(K)[n]$, which has no point of order 9. So $\Phi_n(R_9) \neq 0$; and $\Phi_n(R_9) \in (\mathrm{Res}_{K/\mathbb{Q}} E)(\mathbb{Q}) = E(K)$, where the 3-primary torsion is $\mathbb{Z}/3$. Hence $\Phi_n(R_9) = (P_1, \sigma P_1)$ with $P_1 \in E(K)$ of order 3. If $9 \mid n$, then $\varphi_n(P_1, \sigma P_1) = \varphi_n \Phi_n(R_9) = nR_9 = 0$, so $\psi_n(P_1) = \sigma(P_1)$, against (1). Therefore

$$v_3(n) = 1. \tag{2}$$

The conjugate $(3, 3)$-gluing. Write $n = 3m$ with $3 \nmid m$. The 3-primary part Γ_3 of $\ker \varphi_n$ is a characteristic subgroup of a subgroup defined over $\mathbb{Q}$, hence is defined over $\mathbb{Q}$; by (2), over K it is the graph of $\psi = \psi_n|_{E[3]} : E[3] \to F[3]$. Let $B = \mathrm{Res}_{K/\mathbb{Q}}(E)/\Gamma_3$ with quotient map φ, and let $\Phi : B \to \mathrm{Res}_{K/\mathbb{Q}} E$ be the isogeny with $\Phi \circ \varphi = [3]$, which exists because Γ_3 is killed by 3. Both are defined over $\mathbb{Q}$. The map ψ is an anti-isometry: if $m'm \equiv 1 \pmod 3$, then $e_3(x, y) = e_n(m'x, y)$ for $x, y \in E[3]$, likewise on F, and ψ commutes with m'. By (1), $\psi(P) = -\sigma(P)$, which is (b). As $\Gamma_3 \subseteq \ker \varphi_n$, we have $\varphi_n = \beta \circ \varphi$ for an isogeny $\beta : B \to A$ over $\mathbb{Q}$ whose kernel $\ker \varphi_n / \Gamma_3$ has exponent m; so there is $\beta' : A \to B$ over $\mathbb{Q}$ with $\beta\beta' = [m]$. The point $R = \beta'(R_9) \in B(\mathbb{Q})$ has order 9, since $\beta(R) = mR_9$ has order 9. Finally, over K, $\ker \Phi = \varphi(E[3] \times F[3]) \cong (E[3] \times F[3])/\Gamma_3$ is identified with $F[3]$ as above, so its group of K-rational points is $F(K)[3]$, of order 3. As for R_9, it follows that $\Phi(R) = (P_1', \sigma P_1')$ with $P_1' \in E(K)$ of order 3. Replacing P by P_1', which also satisfies (b) by (1), gives (c). □

Nothing later uses that B is principally polarized, or the description of $B(\mathbb{Q})_{\mathrm{tors}}$ (it is $\mathbb{Z}/45$: its 3-primary part has order at most $\#F(K)[3] \cdot \#E(K)[3] = 9$, and its prime-to-3 part is that of $E(K)$). Section 5 uses only (a)–(c).

5 Two necessary conditions

Lemma 4. In the situation of Lemma 3, choose a model $E : y^2 + a_1xy + a_3y = x^3$ over K with $P = (0, 0)$. Then $a_3 \neq 0$,

$$N_{K/\mathbb{Q}}(\Delta(E)) \in \mathbb{Q}^{\times 3}, \tag{3}$$

$$a_3/\sigma(a_3) \in K^{\times 3}. \tag{4}$$

Condition (3) uses only Lemma 3(b).

Such a model exists: P is a flex because $3P = 0$, and moving P to $(0, 0)$ and its tangent to $y = 0$ forces $a_2 = a_4 = a_6 = 0$. The only changes of variables preserving this shape are $(x, y) \mapsto (u^2x, u^3y)$, which multiply a_3 by u^3 and Δ by u^{12}; so (3) and (4) do not depend on the model. The model of E^σ is the conjugate one, with $\sigma(P) = (0, 0)$. By part B of

the certificate, $\Delta(E) = a_3^3(a_1^3 - 27a_3)$, so $a_3 \neq 0$; the 3-division polynomial is $x\,h(x)$ with $h(x) = 3x^3 + a_1^2x^2 + 3a_1a_3x + 3a_3^2$; and $-P = (0, -a_3)$.

Proof of (3). The extension $0 \to \langle P\rangle \to E[3] \to E[3]/\langle P\rangle \to 0$, with $\langle P\rangle \cong \mathbb{Z}/3$ by $P \mapsto 1$ and $E[3]/\langle P\rangle \cong \mu_3$ by $Q \mapsto e_3(P, Q)$, has a class $c(E)$ in $\mathrm{Ext}^1_{G_K}(\mu_3, \mathbb{Z}/3) = H^1(K, \mathrm{Hom}(\mu_3, \mathbb{Z}/3)) = H^1(K, \mu_3) = K^\times/K^{\times 3}$; here $\mathrm{Hom}(\mu_3, \mathbb{Z}/3) \cong \mu_3$ canonically, as $\mathbb{F}_3^\times = \{\pm 1\}$. We use additive notation in $K^\times/K^{\times 3}$.

(a) $c(E) = \pm[\Delta(E)]$. Over $K(\zeta_3)$, $c(E)$ is the homomorphism describing the unipotent action on $E[3]$, so $K(E[3]) = K(\zeta_3, \sqrt[3]{c(E)})$. The roots of h, the x-coordinates of the points of $E[3] \setminus \langle P\rangle$, are the numbers $x_k = -(e_k^2 + a_1e_k + a_1^2)/9$, where $e_k = \zeta_3^k e_0$ runs through the cube roots of $a_1^3 - 27a_3 = \Delta/a_3^3$ (part B of the certificate); they are distinct, since $x_0 - x_1 = -e_0(1 - \zeta_3)\big(e_0(1 + \zeta_3) + a_1\big)/9$ vanishes only if $e_0^3 \in \{0, a_1^3\}$, that is, if $\Delta = 0$. Conversely $x_0 + \zeta_3 x_1 + \zeta_3^2 x_2 = -e_0^2/3$, so $e_0 = e_0^3/e_0^2$ lies in $K(\zeta_3, x_0, x_1, x_2)$. So $K(E[3])$ contains $K(\zeta_3, \sqrt[3]{\Delta}) = K(\zeta_3, e_0)$, and it is generated over it by y-coordinates, which satisfy quadratic equations; as $[K(E[3]) : K(\zeta_3)]$ is 1 or 3, the two fields coincide. By Kummer theory $c(E)$ and $[\Delta]$ generate the same subgroup of $K(\zeta_3)^\times/K(\zeta_3)^{\times 3}$, hence of $K^\times/K^{\times 3}$, the restriction map being injective as $[K(\zeta_3) : K] \leq 2$.

(b) $c(E^\sigma) = \sigma(c(E))$, by transport of structure under $\tilde\sigma$, which maps E, P, $E[3]$ and e_3 to E^σ, $\sigma(P)$, $E^\sigma[3]$ and e_3; and $\Delta(E^\sigma) = \sigma(\Delta(E))$.

(c) $c(E^\sigma) = -c(E)$. By (1), ψ maps $\langle P\rangle$ to $\langle \sigma P\rangle$ by $\alpha = -1$; it induces on the quotients, identified with μ_3, multiplication by the exponent β with $e_3(\sigma P, \psi Q) = e_3(P, Q)^\beta$. As ψ is an anti-isometry, $e_3(P, Q)^{-1} = e_3(\psi P, \psi Q) = e_3(-\sigma P, \psi Q) = e_3(P, Q)^{-\beta}$, so

$$\beta = +1.$$

An isomorphism of extensions that is α on the subgroups and β on the quotients gives $\alpha\, c(E) = \beta\, c(E^\sigma)$, so $c(E^\sigma) = -c(E)$.

(d) By (a)–(c), $c(E) + \sigma(c(E)) = 0$, that is, $N_{K/\mathbb{Q}}(\Delta)^{\pm 1} \in K^{\times 3}$. Finally $K^{\times 3} \cap \mathbb{Q}^\times = \mathbb{Q}^{\times 3}$: if $q = k^3$ with $k \in K$, then $q^2 = N_{K/\mathbb{Q}}(k)^3$, so $q = q^3/q^2$ is a rational cube. This proves (3).

Proof of (4). Let R, φ, Φ be as in Lemma 3(c), with $\Phi(R) = (P, \sigma P)$. For $U \in E(\bar{\mathbb{Q}})$ with $3U = P$ and $V \in E^\sigma(\bar{\mathbb{Q}})$ with $3V = \sigma P$ we have $\Phi\varphi(U, V) = (3U, 3V)$. So the fibre $\Phi^{-1}(P, \sigma P) = \varphi([3]^{-1}P \times [3]^{-1}\sigma P)$ is a torsor under $\ker\Phi = \varphi(E[3] \times E^\sigma[3])$ with the K-rational point R. Transport it by the G_K-isomorphism $\theta : \ker\Phi \to E^\sigma[3]$, $\varphi(U', V') \mapsto V' - \psi(U')$, which is well defined because $\ker\varphi$ is the graph of ψ. If κ denotes the Kummer map $E(K) \to H^1(K, E[3])$ of multiplication by 3, the class of the transported torsor is $\kappa(\sigma P) - \psi_*\kappa(P) \in H^1(K, E^\sigma[3])$. As the torsor has a K-rational point,

$$\kappa(\sigma P) = \psi_*\kappa(P).$$

Let $\lambda_E : E(K) \to H^1(K, \mu_3) = K^\times/K^{\times 3}$ be κ followed by $E[3] \to E[3]/\langle P\rangle \cong \mu_3$, and define λ_{E^σ} in the same way with σP. Since ψ induces $\beta = +1$ on the quotients, $\lambda_{E^\sigma}(\sigma P) = \lambda_E(P)$; by transport of structure, $\lambda_{E^\sigma}(\sigma P) = \sigma(\lambda_E(P))$. So $\lambda_E(P)$ is fixed by σ.

To compute $\lambda_E(P)$, let $\phi : E \to E' = E/\langle P\rangle$ be the quotient isogeny and $\hat\phi : E' \to E$ its dual, so that $\hat\phi\phi = [3]$. As ϕ maps $[3]^{-1}(Q)$ onto $\hat\phi^{-1}(Q)$ and $E[3]/\langle P\rangle$ isomorphically onto $\ker\hat\phi$, λ_E is the connecting map δ of $0 \to \ker\hat\phi \to E' \to E \to 0$, followed by an isomorphism $\ker\hat\phi \cong \mu_3$. Part B of the certificate checks that $E' : y^2 + a_1xy + a_3y = x^3 - 5a_1a_3x - a_1^3a_3 - 7a_3^2$, that $\hat\phi\phi = [3]$, and that

$$\hat\phi(x, y) = \Big(\frac{x^3 - 9a_1a_3x - a_1^3a_3 - 27a_3^2}{(3x + a_1^2)^2},\ \Big(\frac{y - 4a_3}{3x + a_1^2}\Big)^3\Big).$$

Thus $y \circ \hat{\phi} = w^3$ with $w = (y - 4a_3)/(3x + a_1^2)$ in the function field of E'. For $S \in \ker \hat{\phi}$ the function $w(\cdot + S)/w$ has cube 1, hence is a constant $\xi(S) \in \mu_3$, and $\xi : \ker \hat{\phi} \to \mu_3$ is a homomorphism; it is injective, since otherwise $w = w_0 \circ \hat{\phi}$ with $w_0^3 = y$, which is impossible because $\mathrm{div}(y) = 3(P) - 3(O)$ and $(P) - (O)$ is not principal. For $Q \in E(K) \setminus \{O, P\}$ and Q' with $\hat{\phi}(Q') = Q$ we get $w(Q')^3 = y(Q) \in K^\times$ and $g(w(Q'))/w(Q') = \xi(gQ' - Q')$ for $g \in G_K$, so $\xi_* \delta(Q) = [y(Q)]$. With $Q = -P$ and $y(-P) = -a_3$ this gives $\lambda_E(-P) = \pm[a_3]$ (two isomorphisms $\ker \hat{\phi} \cong \mu_3$ differ by a sign, and -1 is a cube), and $\lambda_E(P) = -\lambda_E(-P) = \mp[a_3]$. As $\lambda_E(P)$ is fixed by σ, so is $[a_3]$, which is (4). □

The sign $\beta = +1$ is where the minus sign in (1) enters: with $\beta = -1$ the argument would give $N_{K/\mathbb{Q}}(a_3) \in K^{\times 3}$ instead of (4).

Put $S = \mathrm{Tr}_{K/\mathbb{Q}}(a_3)$ and $N = N_{K/\mathbb{Q}}(a_3) \neq 0$. If $w \in K$ satisfies $w^3 = a_3/\sigma(a_3)$, then $N_{K/\mathbb{Q}}(w)^3 = 1$, so $N_{K/\mathbb{Q}}(w) = 1$ as it is rational, and $r = w + \sigma(w) \in \mathbb{Q}$ satisfies, since $w\sigma(w) = 1$,

$$r^3 - 3r = w^3 + \sigma(w)^3 = \frac{a_3}{\sigma(a_3)} + \frac{\sigma(a_3)}{a_3} = \frac{S^2}{N} - 2. \tag{5}$$

If $v_\ell(r^3 - 3r) < 0$ for a prime ℓ, then $v_\ell(r) < 0$, so $v_\ell(r^3) = 3v_\ell(r) < v_\ell(3r)$ and

$$v_\ell(r^3 - 3r) = 3v_\ell(r) \equiv 0 \pmod 3. \tag{6}$$

6 A primitive cyclotomic equation

Proposition 5. Let a, b be coprime integers with $\Phi_{10}(a, b) = a^4 - a^3b + a^2b^2 - ab^3 + b^4 \in \{c^3, 5c^3\}$ for some integer c. Then $(a : b)$ is one of $(0 : 1)$, $(1 : 0)$, $(1 : 1)$, $(-1 : 1)$.

Proof. The descent follows Poonen [Po, Section 4], who applies it to $x + \zeta y$ for a solution of $x^5 + y^5 = z^3$ ([Po, (15), p. 198, and p. 199]). Let $L = \mathbb{Q}(\zeta)$ with $\zeta = \zeta_5$, and $\tau = -(\zeta^2 + \zeta^3) = (1 + \sqrt{5})/2$. The ring of integers of L is $\mathbb{Z}[\zeta]$, and its Minkowski bound $(4!/4^4)(4/\pi)^2\sqrt{125} \approx 1.699$ is less than 2, so L has class number one. Its unit group is $\mu_{10} \times \tau^{\mathbb{Z}}$: for a unit u, $u/\bar{u}$ is an algebraic integer all of whose conjugates have absolute value 1, hence a root of unity, and it is $\equiv 1$ modulo $\pi = 1 - \zeta$, so it lies in μ_5; choosing $\eta \in \mu_5$ with $\eta^2 = u/\bar{u}$, the unit u/η is real, hence $\pm\tau^k$.

Put $\alpha = a + b\zeta$. Then $N_{L/\mathbb{Q}}(\alpha) = \Phi_{10}(a, b)$, a positive integer. If a prime $\mathfrak{p}$ divides α and a conjugate $a + b\zeta^k$ with $k \neq 1$, then it divides $b(\zeta - \zeta^k)$, which is $b\pi$ times a unit; if $\mathfrak{p} \neq (\pi)$, it divides b, hence a, which is impossible. So the conjugates of α are pairwise coprime away from π. Now let $\mathfrak{p} \nmid \pi$ divide α, above the prime $\ell \neq 5$. No other conjugate of $\mathfrak{p}$ divides α: if $g\mathfrak{p} \mid \alpha$ with $g\mathfrak{p} \neq \mathfrak{p}$, then $\mathfrak{p}$ divides both α and $g^{-1}\alpha$. For the same reason no $g \neq 1$ fixes $\mathfrak{p}$, so ℓ splits completely, the four primes above ℓ are the conjugates of $\mathfrak{p}$, and $v_\mathfrak{p}(\alpha) = v_\ell(\Phi_{10}(a, b)) \equiv 0 \pmod 3$. At π: $\alpha = (a + b) - b\pi$, so $v_\pi(\alpha) = 0$ if $5 \nmid a + b$, and $v_\pi(\alpha) = 1$ if $5 \mid a + b$ (then $5 \nmid b$ and $v_\pi(a + b) \geq v_\pi(5) = 4$). By class number one, $\alpha = \varepsilon\pi^i w_0^3$ with ε a unit, $i \in \{0, 1\}$ and $w_0 \in L$. Roots of unity are cubes, since $\gcd(3, 10) = 1$, and

$$\pi^4 = 5\zeta^2\tau^{-2}, \qquad \pi = 5\tau\Big(\frac{\zeta^4}{\tau\pi}\Big)^3,$$

so

$$\alpha = 5^i\tau^j w^3 \qquad \text{with } i \in \{0, 1\},\ j \in \{0, 1, 2\},\ w \in L^\times. \tag{7}$$

Modulo 2. The prime 2 is inert in L, with residue field $\mathbb{F}_{16}$, whose multiplicative group is cyclic of order 15; its cubes are its elements of order dividing 5. As a, b are coprime, α is a 2-adic unit, hence so is w. The reduction $\bar{\tau}$ is a root of $X^2 + X + 1$, of order 3, so it is not a cube, and $\bar{5} = 1$. If exactly one of a, b is odd, then $\bar{\alpha} \in \{1, \bar{\zeta}\}$ is a cube ($\bar{\zeta}^5 = 1$), so $j = 0$.

If both are odd, then $\bar\alpha = 1+\bar\zeta$, and $(1+\bar\zeta)^5 = (1+\bar\zeta^4)(1+\bar\zeta) = \bar\zeta+\bar\zeta^4 = \bar\tau^2 = \bar\tau^5$, so $1+\bar\zeta$ is $\bar\tau$ times a cube and $j=1$. Thus $j\in\{0,1\}$.

Two curves over L. The automorphisms $\zeta\mapsto\zeta^2$ and $\zeta\mapsto\zeta^3$ map τ to $-\tau^{-1}$. Multiplying (7) and its images under these two automorphisms gives $(a+b\zeta)(a+b\zeta^2)(a+b\zeta^3) = 5^{3i}\tau^{-j}W^3$ with $W\in L$. If $b\neq 0$, dividing by b^3 shows that $t=a/b$ is the first coordinate of an L-point of

$$D_j:\quad Y^3 = \tau^j(t+\zeta)(t+\zeta^2)(t+\zeta^3),\qquad j\in\{0,1\};\tag{8}$$

the factor 5^{3i} is a cube. The case $b=0$, that is $(a:b)=(1:0)$ with $j=0$, is the point $(t:Y:W)=(1:1:0)$ at infinity of the projective model $Y^3=\tau^j(t+\zeta W)(t+\zeta^2W)(t+\zeta^3W)$. Each D_j is a smooth plane cubic with the L-rational flex $(-\zeta,0)$, hence an elliptic curve. With this point as origin, D_j is isomorphic over L to

$$\begin{aligned} E_0:&\ y^2+(4\zeta^3+3\zeta^2+7\zeta+6)y = x^3-10\zeta^3-5\zeta^2-10\zeta,\\ E_1:&\ y^2+(\zeta^3+7\zeta^2+13\zeta+9)y = x^3-35\zeta^3-55\zeta^2-35\zeta. \end{aligned}$$

Part C of the certificate proves: $N_{L/\mathbb{Q}}(\Delta(E_j)) = 3^{12}5^{12}$; E_j has rank 1 and $E_j(L)\cong\mathbb{Z}/3\oplus\mathbb{Z}$ with proved generators (Magma's `RankBounds` and `MordellWeilGroup`, with class groups computed unconditionally up to the Minkowski bound); and elliptic Chabauty [Br] at the prime 11, which splits completely in L and is a prime of good reduction, returns an upper bound equal to the number of points found and no unresolved residue classes. The saturation it requires (at 2 for $j=0$, none for $j=1$) holds because the Mordell–Weil groups are proved to be complete. The L-points of D_j with rational t-coordinate are therefore exactly those with

$$t\in\{0,\infty\}\ \text{ on } D_0,\qquad t\in\{1,-1\}\ \text{ on } D_1,\tag{9}$$

which gives the four values of $(a:b)$. □

In [Po, Section 4] the descent (7) is followed by an analysis of the curves $(U+1)(U^5+1)=\eta V^3$ and their elliptic quotients over $\mathbb{Q}(\sqrt5)$. Here the equation is different (the norm $\Phi_{10}(a,b)$ itself is a cube or five times a cube); what is new is that the product of three conjugates removes the factor 5^{3i}, so that only the two cubic twists D_0, D_1 remain, and these are treated by elliptic Chabauty over $\mathbb{Q}(\zeta_5)$.

7 The quadratic points of $X_1(15)$

Proposition 6. Let K be a quadratic field, E an elliptic curve over K with $E(K)_{\mathrm{tors}}\cong\mathbb{Z}/15$, and $P\in E(K)$ a point of order 3. Write $E:\ y^2+a_1xy+a_3y=x^3$ with $P=(0,0)$. Then (3) and (4) do not both hold.

A model of $X_1(15)$. Let $P_5\in E(K)$ have order 5. By (I9), $(E,P_5)\cong(E_d,(0,0))$ over K for some $d\in K$ with $d^5(d^2-11d-1)\neq 0$; let (x_3,y_3) be the image of P. The 3-division polynomial of E_d is

$$\psi_3(X) = 3X^4+(d^2-6d+1)X^3+3(d^2-d)X^2+3d^2X-d^3,$$

and $\psi_3(0)=-d^3\neq0$, $\psi_3(d)=d^5\neq0$. So $u=x_3/d$ is neither 0 nor 1, and with $z=du/(u-1)$ one has $\psi_3(du)/d^3=(u-1)^2\big(uz^2+3u(u-1)z+(u-1)\big)$ (part D of the certificate). Hence (u,z,y_3) is a K-point of the affine curve X^{aff} given by

$$\begin{aligned} &uz^2+3u(u-1)z+(u-1)=0,\qquad d=\frac{(u-1)z}{u},\qquad x_3=(u-1)z,\\ &y_3^2+(1-d)x_3y_3-dy_3=x_3^3-dx_3^2, \end{aligned}\tag{10}$$

where the last equation, multiplied by u^2, is a polynomial equation in u, z, y_3. The point $(0,0)$ of E_d has order 5 and (x_3, y_3) has order 3. Translating (x_3, y_3) to $(0,0)$ and shearing its tangent to $y = 0$ turns E_d into a model as in Proposition 6, with

$$a_3 = 2y_3 + (1-d)x_3 - d, \qquad \delta = d^2(d^2 - 11d - 1); \tag{11}$$

the translation contributes $2y_3 + (1-d)x_3$ to the coefficient of y and the shear does not change it, and the discriminant $d^5(d^2 - 11d - 1)$ is unchanged and differs from δ by the cube d^3. So, by the remarks after Lemma 4, (3) and (4) are conditions on δ and a_3 at the point (u, z, y_3). We call a point of (the projective closure of) X^{aff} a *cusp* if d is 0 or ∞ or a root of $d^2 - 11d - 1$ there; the point (u, z, y_3) is not a cusp. The curve X^{aff} is irreducible, and its singular points are $(1, 0, 0)$, where $d = 0$, and two conjugate points with $d^2 - 11d - 1 = 0$ (part D). So (u, z, y_3) is a smooth point of X^{aff}.

The curve W. Let $W:\ y^2 + xy + y = x^3 + x^2$ (Cremona 15a8), with origin O and the points $T_1 = (0,0)$, $T_2 = (-1,0) = 2T_1$, $T_3 = (0,-1) = 3T_1$. It has rank 0 and $W(\mathbb{Q}) = \{O, T_1, T_2, T_3\} \cong \mathbb{Z}/4$ (part D). The rational functions

$$u = \frac{x+1}{x^2+x+1}, \qquad z = \frac{x\,(x^2 - 1 - (x+2)y)}{(x+1)(x^2+x+1)},$$
$$y_3 = \frac{x^5\big(x - 2x^2 - 2x^3 + (x^2+4x+1)y\big)}{(x+1)^2(x^2+x+1)^3}$$

on W define a birational map $W \to X^{\mathrm{aff}}$, with inverse

$$x = \frac{(u-1)^3(u+1)^2 + u^2(u-1)^2 z + u^2(-9u^3 + 3u^2 + 7u + 1)y_3 - u^3(3u+2)zy_3}{u(u-1)^2(u^2 - u - 1)},$$
$$y = -\frac{(u-1)^4 + (u-1)^3 z + 3u^2(u-1)y_3 + u^2 z y_3}{u(u-1)^3}$$

(part D checks that the image lies on (10) and that the composite $W \to X^{\mathrm{aff}} \to W$ is the identity). On W,

$$d = \frac{x^3\big(1 - x^2 + (x+2)y\big)}{(x+1)^2(x^2+x+1)}, \qquad x_3 = du;$$

d has poles at O and T_2 and zeros at T_1 and T_3, so the four rational points of W are cusps. As (u, z, y_3) is a smooth point of X^{aff}, there is a unique point $Q \in W(K)$ at which the functions u, z, y_3 take these values, and the functions a_3 and δ of (11) are regular at Q. Since Q is not a cusp, $Q \notin W(\mathbb{Q})$, so Q is a quadratic point with residue field K, with conjugate Q^σ.

The four families. The divisor $(Q)+(Q^\sigma)$ is rational of degree 2, so it is linearly equivalent to $(O)+(T)$ with $T = Q + Q^\sigma \in W(\mathbb{Q})$. For each $T \in W(\mathbb{Q})$, by Riemann–Roch, the effective divisors linearly equivalent to $(O) + (T)$ are the fibres of a morphism $W \to \mathbb{P}^1$ of degree 2, unique up to an automorphism of $\mathbb{P}^1$; we fix it as the function t_j below, whose divisor of poles lies in this class. So $(Q) + (Q^\sigma)$ is the fibre of some t_j over a point $t_0 \in \mathbb{P}^1(\mathbb{Q})$; as Q is not a cusp and the poles of t_j are rational points, $t_0 \in \mathbb{Q}$.

(1) The class of $(O) + (T_2)$: $t_1 = y/(x + y + 1)$, with divisor $2(T_1) - 2(T_3)$ and $t_1(O) = t_1(T_2) = 1$. Over $\mathbb{Q}(t_1)$, $(1-t_1)^2x^2 - t_1 x - t_1 = 0$ and $y = t_1(x+1)/(1-t_1)$.

(2) The class of $2(O)$: $t_2 = x + 1$, with divisor $2(T_2) - 2(O)$ and $t_2(T_1) = t_2(T_3) = 1$. Over $\mathbb{Q}(t_2)$, $y^2 + t_2 y - t_2(t_2 - 1)^2 = 0$ and $x = t_2 - 1$.

(3) The class of $(O) + (T_3)$: $t_3 = y/x$, with divisor $(T_1) + (T_2) - (T_3) - (O)$, normalized by $t_3 = -1$ at the cusp $(\omega, -\omega)$ of degree 2, where $\omega^2 + \omega + 1 = 0$. Over $\mathbb{Q}(t_3)$, $x^2 + (1 - t_3 - t_3^2)x - t_3 = 0$ and $y = t_3 x$.

(4) The class of $(O)+(T_1)$: $t_4=-(x+y+1)/x$, with divisor $(T_2)+(T_3)-(T_1)-(O)$, normalized by $t_4=-1$ at the cusp $(\omega,-1)$. Over $\mathbb{Q}(t_4)$, $x^2+(1-t_4-t_4^2)x-t_4=0$ and $y=-(t_4+1)x-1$.

Each t_j is determined by its divisor and its normalization, so the families do not depend on any choice (part D checks the divisors and values). The negation of W exchanges families 3 and 4 and maps t_1 to $1/t_1$.

Traces and norms. In each family the function field $\mathbb{Q}(W)$ is the quadratic extension of $\mathbb{Q}(t)$, $t=t_j$, given above. Computing in it (part D), the function $R_j=\mathrm{Tr}(a_3)^2/N(a_3)-2\in\mathbb{Q}(t)$, formed with the trace and norm from $\mathbb{Q}(W)$ to $\mathbb{Q}(t)$, is $R_j=N_j(t)/D_j(t)$, where, with $B(t)=t^2-t+1$, $\Phi_{10}(t)=t^4-t^3+t^2-t+1$ and $\Phi_5(t)=t^4+t^3+t^2+t+1$,

$$D_1=D_2=B^3\Phi_{10},\qquad D_3=D_4=t^9\Phi_5,$$

$$\begin{aligned}
N_1&=-2t^{10}-17t^9+60t^8-30t^7-126t^6+228t^5-126t^4-30t^3+60t^2-17t-2,\\
N_2&=2t^{10}+8t^9-24t^8+30t^7-45t^6+60t^5-45t^4+30t^3-24t^2+8t+2,\\
N_3&=-t^{22}-6t^{21}-7t^{20}+16t^{19}+21t^{18}-44t^{17}-37t^{16}+88t^{15}+38t^{14}-132t^{13}-17t^{12}\\
&\quad+152t^{11}-17t^{10}-132t^9+38t^8+88t^7-37t^6-44t^5+21t^4+16t^3-7t^2-6t-1,\\
N_4&=-t^{22}-5t^{21}-2t^{20}+15t^{19}-8t^{18}-37t^{17}+55t^{16}+20t^{15}-119t^{14}+89t^{13}+73t^{12}\\
&\quad-170t^{11}+73t^{10}+89t^9-119t^8+20t^7+55t^6-37t^5-8t^4+15t^3-2t^2-5t-1.
\end{aligned}$$

The N_j have integer coefficients, are palindromic, and

$$\mathrm{Res}(\Phi_{10},N_1)=\mathrm{Res}(\Phi_{10},N_2)=\mathrm{Res}(\Phi_5,N_3)=\mathrm{Res}(\Phi_5,N_4)=25.$$

Moreover the norm of δ from $\mathbb{Q}(W)$ to $\mathbb{Q}(t)$ is $F_j(t)\,g_j(t)^3$ with $g_j\in\mathbb{Q}(t)$ (computed in part D) and

$$\begin{aligned}
F_1&=F_2=(t^2-t+1)(t^4-11t^3+21t^2-11t+1),\\
F_3&=F_4=(t^2-t+1)^2(t+1)(t^4+t^3-9t^2+t+1).
\end{aligned}$$

Specialization. Let $\mathcal{O}$ be the integral closure of the local ring $\mathbb{Q}[t]_{(t-t_0)}$ in $\mathbb{Q}(W)$. As $\{Q,Q^\sigma\}$ is the whole fibre of t over t_0, $\mathcal{O}$ is the local ring of W at this point of degree 2; it is free of rank 2 over $\mathbb{Q}[t]_{(t-t_0)}$, with $\mathcal{O}/(t-t_0)\mathcal{O}\cong K$, and $a_3,\delta\in\mathcal{O}$. Traces and norms are traces and determinants of multiplication maps in a basis, so they commute with reduction modulo $t-t_0$: with S and N as in (5), $S^2/N-2=R_j(t_0)$, and $N_{K/\mathbb{Q}}(\delta(Q))=F_j(t_0)g_j(t_0)^3$ whenever $g_j(t_0)$ is finite and nonzero.

The arithmetic. Suppose that (4) holds. By (5), $r^3-3r=R_j(t_0)$ for some $r\in\mathbb{Q}$. Write $t_0=a/b$ with coprime integers a, b, $b>0$. In families 1 and 2, $R_j(t_0)=N_j(a,b)/\big(B(a,b)^3\Phi_{10}(a,b)\big)$, a quotient of binary forms of degree 10. Let $\ell\neq5$ be a prime dividing $\Phi_{10}(a,b)$. Then $\ell\nmid b$ (otherwise $\ell\mid a$), and $\ell\nmid N_j(a,b)$: the resultant of the binary forms Φ_{10} and N_j is $\mathrm{Res}(\Phi_{10},N_j)=25$, as both have full degree in t, and it times a power of a and times a power of b lie in the ideal generated by $\Phi_{10}(a,b)$ and $N_j(a,b)$. So $v_\ell(R_j(t_0))=-3v_\ell(B(a,b))-v_\ell(\Phi_{10}(a,b))<0$, and (6) gives $v_\ell(\Phi_{10}(a,b))\equiv0\pmod3$. Also $\Phi_{10}(a,b)=N_{L/\mathbb{Q}}(a+b\zeta)>0$ and $v_5(\Phi_{10}(a,b))=v_\pi(a+b\zeta)\in\{0,1\}$ (Section 6). Hence $\Phi_{10}(a,b)$ is a cube or five times a cube, and Proposition 5 gives $t_0\in\{0,\infty,1,-1\}$. In families 3 and 4, if $t_0\neq0$, then $R_j(t_0)=N_j(a,b)/\big(a^9b^9\Phi_5(a,b)\big)$, a quotient of forms of degree 22; a prime $\ell\neq5$ dividing $\Phi_5(a,b)$ divides neither a, b nor $N_j(a,b)$, so in the same way $\Phi_5(a,b)=\Phi_{10}(-a,b)$ is a cube or five times a cube, and Proposition 5 for $(-a,b)$ gives the same four values of t_0.

The four fibres. Part D determines the fibres of t_j over 0, 1, -1 and ∞. Three of them consist of cusps, and the fourth, over t_0, is one non-cuspidal point with residue field K, where

d is rational:

j	fibres of cusps	t_0	K	d	$N_{K/\mathbb{Q}}(\Delta) \bmod \mathbb{Q}^{\times 3}$
1	$0,\ 1,\ \infty$	-1	$\mathbb{Q}(\sqrt{-15})$	$-\frac{1}{8}$	$F_1(-1) = 135$
2	$0,\ 1,\ \infty$	-1	$\mathbb{Q}(\sqrt{-15})$	8	$F_2(-1) = 135$
3	$0,\ -1,\ \infty$	1	$\mathbb{Q}(\sqrt{5})$	$\frac{1}{2}$	$F_3(1) = -10$
4	$0,\ -1,\ \infty$	1	$\mathbb{Q}(\sqrt{5})$	-2	$F_4(1) = -10$

At the non-cuspidal point over t_0 the curve E_d has $E_d(K)_{\mathrm{tors}} \cong \mathbb{Z}/15$, the cube cofactor $g_j(t_0)$ is finite and nonzero (it is $5/768$, $80/3$, $-5/8$, -10 in families 1–4), and $N_{K/\mathbb{Q}}(\Delta(E_d))$ is 135 resp. -10 times a rational cube. Neither 135 nor -10 is a rational cube, so (3) fails there. Hence (3) and (4) never hold together. □

8 Proof of the Theorem and consequences for the census

Proof of the Theorem. Let $A/\mathbb{Q}$ be principally polarized, simple over $\mathbb{Q}$ and geometrically split, with a rational point of order 45. By (I3), A is of type (N), (CM) or (S). Types (CM) and (S) are excluded by Proposition A. In type (N), Lemma 3 gives a quadratic field K, an elliptic curve E over K with $E(K)_{\mathrm{tors}} \cong \mathbb{Z}/15$ and a point $P \in E(K)$ of order 3, and Lemma 4 shows that (3) and (4) hold for the model of E with $P = (0, 0)$. This contradicts Proposition 6. □

Proof of the Corollary. The Jacobian of a curve of genus 2 over $\mathbb{Q}$ is a principally polarized abelian surface, and a group containing an element of order 45 contains a cyclic subgroup of order 45. □

Among the groups listed on the census, [45] is the only one with an element of order 45. It becomes impossible in the class "geometrically split, simple over $\mathbb{Q}$". Simplicity over $\mathbb{Q}$ is essential: the class "split over $\mathbb{Q}$" contains the curve 45.a and a certified infinite family of Jacobians with exact torsion group [45] (see the census page), so the combined geometrically split cell of [45] keeps its grade. The class "geometrically simple" is not affected by this proof.

9 Remarks

1. *The hypotheses.* The principal polarization and the simplicity over $\mathbb{Q}$ are used only in Section 4. Simplicity makes the two elliptic curves conjugate. The principal polarization identifies A_K/F with E itself, so that the torsion of G/H lands in $E(K)$, and makes A_K an (n, n)-gluing with anti-isometric kernel; for a polarization of higher degree A_K/F is only isogenous to E. Proposition A uses neither hypothesis; Lemma 2 and Propositions 5 and 6 are statements about elliptic curves and a Diophantine equation; and Lemma 4 uses only the output (a)–(c) of Lemma 3.

2. *Both conditions are needed.* At the non-cuspidal points of the table in Section 7, condition (4) holds ($a_3/\sigma(a_3) = -1$, so $r = -2$ and $R_j(t_0) = -2$), and only (3) excludes them. There d is rational: the curves are the base changes to $\mathbb{Q}(\sqrt{-15})$ of the curve 50b2 ($d = -\frac{1}{8}$, 8) and to $\mathbb{Q}(\sqrt{5})$ of the curve 50b1 ($d = \frac{1}{2}$, -2), so $E \cong E^\sigma$. Conversely, at the point of family 1 over $t_1 = 2$, over $\mathbb{Q}(\sqrt{3})$, condition (3) holds and (4) fails; this point gives the elliptic curve of the census curve 15.c (Remark 3).

3. *A consistency check with census curves (not used in the proof).* The census curves 9.c, 18.c and 15.c are geometrically split and simple over $\mathbb{Q}$, with rational torsion

subgroups [9], [18] and [15]. The census records for each a map of degree 3 to an elliptic curve E over $K = \mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{6})$ and $\mathbb{Q}(\sqrt{3})$. As 3 is prime, the map does not factor through an isogeny, so the induced homomorphism $E \to J_K$ is injective [BL, Proposition 11.4.3]. Part E of the certificate checks that E has no complex multiplication and is not geometrically isogenous to E^σ (the squares of the traces of Frobenius at two conjugate primes differ), so these Jacobians are of type (N) and E is one of the two elliptic curves of Section 4; and that $E(K)_{\mathrm{tors}}$ is $\mathbb{Z}/3$, $\mathbb{Z}/6$ and $\mathbb{Z}/15$. For 9.c and 18.c the argument of Section 4, with the subgroup of order 9 in place of the group of order 45, forces $d = 3$ and hence (1), (3) and (4); and indeed (3) and (4) hold for both. For 15.c, which has no rational point of order 9, (3) holds and (4) fails. The variant $a_3\sigma(a_3) \in K^{\times 3}$ of (4) with the opposite sign in (1) fails for all three curves.

4. *What was not used.* The proof uses no bound on the height of points, no search for points as a completeness argument and no conjecture: the ranks, the Mordell–Weil groups and the Chabauty bounds of Section 6 and the rank of W are proved.

10 Computations

The certificate `qsimple_45_checks.m` (Magma V2.30-1, about 11 seconds) consists of assertions and ends with `ALL_CHECKS_PASSED`; a failed assertion stops it before that line. Its log is `qsimple_45_checks.log`. SHA-256 of the script: `3898e5d57d639cd43555feb97c996164ef867549fd76b5df02bf40c2da0a0fe6`; of the log: `4681daf8118e304a60d91375822b1a8874028e922ed94e4ea63803f907b3f060`. Part A checks the finite statements of (I1), the residues modulo 5 and 3 used in Section 3 and $(1+\sqrt{2})^4 < 34$. Part B checks, for $y^2 + a_1xy + a_3y = x^3$ over $\mathbb{Q}(a_1, a_3)$, the discriminant, the 3-division polynomial, $-P$, the roots of h, the curve E' and the formula for $\hat{\phi}$ of Section 5, including $\hat{\phi}\phi = [3]$. Part C proves the statements of Section 6: the Minkowski bound, class number one, the unit group, the action of $\zeta \mapsto \zeta^2, \zeta^3$ on τ, the norm form, the identities for π, the cube classes modulo 2; for $j = 0, 1$ that D_j is smooth of genus 1 and isomorphic to E_j, the norm of $\Delta(E_j)$, the splitting of 11, the rank bounds, the Mordell–Weil groups with both proof flags, the map $E_j \to D_j \to \mathbb{P}^1$, and that Chabauty at 11 returns a bound equal to the number of points found, no unresolved residue classes, the saturation index and the sets (9). Part D treats Section 7: the Tate normal form, ψ_3 and the identity for $\psi_3(du)/d^3$; the irreducibility of (10) and its singular points; the rank and torsion of W; the birational map and its inverse; the identities (11) by an explicit translation and shear; the cusps; the divisors of the t_j; in each family the quadratic relation, the identity $R_j = N_j/D_j$, the integrality, degrees and palindromy of N_j, the resultants, and an explicit cube root g_j of $N(\delta)/F_j$ (the leading coefficient is kept, as Magma's factorization drops it); and the fibres over 0, 1, -1, ∞, including the residue field, d, the torsion, $a_3/\sigma(a_3)$, $N_{K/\mathbb{Q}}(\Delta)$ and $g_j(t_0)$ at the non-cuspidal points. Part E carries out the checks of Remarks 2 and 3 from the census data of 9.c, 18.c and 15.c. All computations were rerun, and the new identities of Sections 5 and 7, the exceptional fibres and the arithmetic of Proposition 5 were cross-checked independently in Sage and PARI.

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